

SIMATS ENGINEERING

Department of Computer Science and Engineering

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Comparative Analysis
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	Moniga V	Reg. No.:	192521184
Section / Batch:	2025 - 2029	Date:	25/08/2026

Instructions to Students

- Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
- Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption.
- Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
- Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable.
- Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

QUESTIONS

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.
2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.
3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.
4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.
5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess their performance advantages and limitations.

Code of Conduct

I Moniga V(192521184) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Moniga V

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Marks Evaluation:

Q. No	Scope of Items (20)	Comparison Depth (25)	Use of Evidence (20)	Critical Thinking (20)	Clarity of Presentation (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	

3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

Question 1: L1, L2, and L3 Cache Levels — Latency, Bandwidth, Throughput

Modern processors use a multi-level cache hierarchy to bridge the growing speed gap between the CPU core and main memory (DRAM). Each level trades capacity against speed: as we move from L1 to L3, capacity increases while speed decreases, because larger, cheaper memory arrays inherently take longer to access.

Comparative Characteristics

Parameter	L1 Cache	L2 Cache	L3 Cache
Location	Inside each core	Per core or per core-pair	Shared across all cores
Typical Size	16–64 KB (split I/D)	256 KB–1 MB	8–64 MB
Access Latency	~1–4 cycles (fastest)	~10–20 cycles	~30–70 cycles
Bandwidth	Highest (multiple ports, wide bus)	High, but lower than L1	Moderate; shared among cores
Throughput	Very high per-core instruction/data supply	Supports L1 misses efficiently	Reduces traffic to main memory
Technology	SRAM	SRAM	SRAM (sometimes eDRAM)
Cost per bit	Highest	High	Lower than L1/L2, higher than DRAM

Analysis of Influence on Processor Performance

L1 cache is split into separate instruction and data caches placed directly beside the execution units, so it must respond within a single-digit cycle count to avoid stalling the pipeline; its small size keeps hit latency minimal but also limits its hit rate for large working sets.

L2 cache acts as a victim/inclusion buffer for L1 misses. Its larger capacity absorbs a significant fraction of L1 misses at a moderate latency penalty, preventing many requests from having to travel all the way to L3 or DRAM.

L3 cache, being shared, allows inter-core data sharing (useful for multi-threaded workloads) and further filters memory traffic. Although its latency is much higher than L1/L2, it is still an order of magnitude faster than DRAM, so it substantially reduces Average Memory Access Time (AMAT) for the system as a whole.

Effect on AMAT

$AMAT = Hit_Time(L1) + Miss_Rate(L1) \times [Hit_Time(L2) + Miss_Rate(L2) \times (Hit_Time(L3) + Miss_Rate(L3) \times Memory_Access_Time)]$. Because each successive level has a much higher hit rate than a direct DRAM access

alone, the multi-level hierarchy keeps effective access time close to L1 speed for most references, even though DRAM latency may be 100+ cycles.

Conclusion

The three-level cache hierarchy provides an optimal balance: L1 maximizes speed for the hottest data, L2 provides a fast second line of defense, and L3 maximizes effective capacity and inter-core sharing before falling back to slow main memory — collectively minimizing AMAT and maximizing sustained instruction throughput.

Question 2: SRAM vs DRAM — Performance and Justification of Use

Comparative Characteristics

Parameter	SRAM	DRAM
Cell Structure	6 transistors (bistable flip-flop)	1 transistor + 1 capacitor
Access Latency	Very low (~1–4 ns)	Higher (~40–80 ns)
Bandwidth	High, due to direct transistor switching	Lower per cell; needs wide buses/burst modes to compensate
Throughput	High — no refresh interruption	Reduced by periodic refresh cycles
Refresh Requirement	None (static)	Needs refresh every ~64 ms (dynamic)
Density / Cost per bit	Low density, expensive	High density, cheap
Power Consumption	Higher static power, low per-access energy	Lower static power, but refresh adds energy overhead
Volatility	Volatile	Volatile

Justification: SRAM for Cache Memory

- Speed-critical role: Cache must match near-CPU-core speeds; SRAM's flip-flop design allows single-cycle access, essential for keeping the pipeline fed without stalls.
- No refresh overhead: Because SRAM does not need periodic refreshing, it delivers consistent, predictable low latency — critical for deterministic pipeline timing.
- Small capacity is acceptable: Caches are intentionally small (KB–MB range), so SRAM's low density and high cost per bit are an acceptable trade-off for the speed gained.

Justification: DRAM for Main Memory

- Capacity requirement: Main memory must hold gigabytes of program and data; DRAM's simple 1T1C cell allows far higher density than SRAM at a fraction of the cost.
- Cost-effectiveness: Large-capacity SRAM would be prohibitively expensive and power-hungry, whereas DRAM's low cost per bit makes multi-GB memory economically feasible.
- Acceptable latency trade-off: Main memory access is already buffered by three cache levels, so DRAM's higher latency is hidden from the CPU most of the time.

Conclusion

The SRAM/DRAM split reflects a classic speed-versus-capacity trade-off in the memory hierarchy: SRAM is reserved for small, extremely fast cache levels where every cycle counts, while DRAM serves as a large, economical main memory whose higher latency is amortized and hidden by the cache hierarchy above it.

Question 3: Virtual Memory, Paging, and Cache Memory Mechanisms

Comparative Overview

Aspect	Cache Memory	Virtual Memory (Paging)
Purpose	Bridges CPU–main memory speed gap	Extends apparent memory size beyond physical RAM
Unit of Transfer	Cache line/block (e.g., 64 bytes)	Page (e.g., 4 KB)
Managed By	Hardware (cache controller)	OS + MMU (hardware-assisted)
Access Latency Impact	Hit: very fast; Miss: fetch from next level	Hit (in RAM): fast; Miss (page fault): very slow disk/SSD access
Address Translation	Tag comparison (associative lookup)	Page table + TLB lookup, virtual-to-physical mapping
Memory Utilization	Exploits temporal/spatial locality of hot data	Allows multiple processes to share physical RAM efficiently via demand paging
Throughput Effect	Reduces effective memory access time (AMAT)	Increases usable address space; excessive faults (thrashing) reduce throughput

How They Contribute to Efficient Memory Management

Cache memory exploits locality of reference: recently and frequently accessed instructions/data are kept close to the CPU, so most references are satisfied at very low latency, directly boosting throughput and reducing AMAT.

Paging-based virtual memory decouples the process's logical address space from limited physical DRAM. The MMU translates virtual addresses to physical addresses via page tables, with a Translation Lookaside Buffer (TLB) caching recent translations to avoid repeated page-table walks. This allows efficient multiprogramming, memory protection, and isolation between processes.

Together, cache and virtual memory form a unified hierarchy: the TLB itself is a cache for address translations, and both mechanisms rely on the principle of locality — cache exploits it at the instruction/data level, while paging exploits it at the working-set level, keeping only actively used pages resident in RAM.

Performance Interpretation

- A high cache hit rate keeps effective access latency close to SRAM speed, improving instruction throughput.
- A low page-fault rate keeps virtual memory overhead negligible; frequent faults (thrashing) can degrade throughput by orders of magnitude since disk/SSD access is far slower than DRAM.
- Well-tuned page size and cache block size both balance the trade-off between transfer overhead and locality exploitation.

Conclusion

Cache memory and paging-based virtual memory operate at different levels of the hierarchy but share the same underlying goal: minimizing effective access latency and maximizing throughput by keeping frequently used data

as close to the processor as possible, while paging additionally provides the illusion of abundant memory and safe multiprogramming.

Question 4: NAND Flash vs DRAM — Suitability in the Memory Hierarchy

Comparative Characteristics

Parameter	DRAM	NAND Flash
Access Latency	Tens of nanoseconds	Tens of microseconds (~1000x slower than DRAM)
Bandwidth	High (GB/s, wide parallel buses)	Moderate (limited by controller and page-based access)
Throughput	High random-access throughput	High sequential throughput; weaker at random small reads/writes
Volatility	Volatile (loses data on power-off)	Non-volatile (retains data without power)
Write Granularity	Byte/word addressable	Page-based writes; block-level erase required before rewrite
Endurance	Effectively unlimited write cycles	Limited program/erase cycles (wear-out over time)
Density / Cost per bit	Moderate density, higher cost per bit	Very high density, low cost per bit
Typical Role	Main memory (working data)	Secondary storage (SSD, persistent storage)

Suitability Analysis

DRAM's nanosecond-scale latency and byte-addressable random access make it ideal as main memory, where the CPU needs frequent, unpredictable, low-latency access to active program data. Its volatility is acceptable because main memory content is reconstructed from storage at boot or reload.

NAND Flash trades speed for non-volatility, density, and cost efficiency. Its block-erase/page-write architecture and microsecond latencies make it unsuitable as a CPU-facing memory, but excellent as secondary storage (SSDs) where persistence across power cycles and low cost per gigabyte matter more than nanosecond latency.

- DRAM sits directly above cache and below the CPU registers/cache hierarchy — fast, volatile, moderate capacity.
- NAND Flash sits at the secondary storage tier — slower, non-volatile, very high capacity, used for persistent data and swap/page storage.
- The latency gap between DRAM and NAND Flash is bridged in some systems by intermediate technologies (e.g., Optane/3D XPoint) or by aggressive caching/prefetching strategies.

Conclusion

DRAM and NAND Flash serve complementary, non-overlapping roles: DRAM's speed justifies its position as main memory despite volatility and higher cost, while NAND Flash's non-volatility, density, and low cost per bit justify its role as persistent secondary storage despite much higher latency.

Question 5: MRAM vs RRAM — Emerging Non-Volatile Memory Technologies

Comparative Characteristics

Parameter	MRAM (Magnetoresistive RAM)	RRAM (Resistive RAM)
Storage Mechanism	Magnetic tunnel junction; stores data via electron spin orientation	Resistance switching in a metal-oxide dielectric layer (formation/rupture of conductive filaments)
Access Time	Very fast (~ns range, close to SRAM)	Fast (~ns to tens of ns), slightly higher variability
Throughput	High, supports fast read/write cycles	High, though switching variability can affect consistency
Energy Efficiency	Low write energy; no leakage since non-volatile	Very low write energy; simple two-terminal structure aids efficiency
Endurance	Very high (10^{12} – 10^{15} cycles), near-DRAM-like	Moderate–high (10^6 – 10^{12} cycles), technology-dependent
Density/Scalability	Moderate density; scaling limited by magnetic film fabrication	Very high potential density; simple structure enables 3D stacking
Non-volatility	Yes	Yes
Maturity	Commercially available (e.g., STT-MRAM)	Emerging, some commercial deployment

Potential Applications in Future Hierarchical Memory Systems

- MRAM: Well suited as a universal memory candidate to replace SRAM/DRAM in cache or main memory tiers, offering near-SRAM speed with non-volatility, enabling instant-on systems and reduced refresh/leakage power.
- RRAM: Strong candidate for high-density storage-class memory (SCM) bridging the gap between DRAM and NAND Flash, and for neuromorphic/in-memory computing due to its analog resistance states.
- Both technologies could enable persistent caches, eliminating the need to reload data from slower storage after power loss, and reduce overall system energy consumption by removing DRAM refresh cycles.

Performance Advantages and Limitations

MRAM's key advantage is its combination of speed and endurance approaching SRAM/DRAM levels while remaining non-volatile, but it faces scaling challenges in achieving very high density comparable to NAND Flash or DRAM.

RRAM's key advantage is its simple, highly scalable structure enabling very high density and 3D integration, along with very low switching energy, but it currently faces variability in resistance states and comparatively lower endurance than MRAM in some implementations, which can affect reliability for write-intensive workloads.

Conclusion

MRAM and RRAM both represent promising steps toward a unified, non-volatile memory hierarchy that could reduce the traditional latency and energy gaps between cache, main memory, and storage. MRAM currently offers superior endurance and speed for cache/main-memory roles, while RRAM offers superior density and scalability, making it attractive for storage-class memory and future in-memory computing architectures — the two together illustrate the ongoing trade-off between speed/endurance and density/cost in next-generation memory design.